Dooho Lee

Co-Founder @ Nums AI

dooho@nums.world

Google Scholar | LinkedIn | GitHub



I am Co-Founder at Nums AI, where we build foundation models for structured data, including tables, graphs, and time series. I am also an M.S. student in Electrical Engineering at KAIST's DataAI Lab, advised by Prof. Jaemin Yoo and Prof. Kijung Shin. Previously, I co-founded startups and worked as a software engineer and designer on products spanning LLM-powered workflows, finance, and mobile applications.

Education

M.S. in Electrical Engineering, KAIST

2025 - Present | Advisor: Prof. Jaemin Yoo and Prof. Kijung Shin | Daejeon, Republic of Korea. GPA: 3.95.

B.S. in Electrical Engineering and Computer Science, KAIST

2019 - 2024 | Double major. Daejeon, Republic of Korea. Cum laude, GPA: 3.77.

Models

Ephris | Nums AI

A graph foundation model for node classification.

arXiv | GitHub | Hugging Face

Causilo | Nums AI

A tabular foundation model for classification and regression.

arXiv | GitHub | Hugging Face

Publications

Structural Domain Adaptation via Operator Alignment

Biniyam Aschalew Tolera, Akotet Yeshaw Tesema, Cheonwoo Lee, **Dooho Lee**, Aikta Arya, Jaemin Yoo

WSDM 2027

Node4All: Learning Node Representation Beyond Datasets

Dooho Lee, Jaemin Yoo

KDD 2026 | GFM @ ICML 2026 | ACM | arXiv | GitHub

View Space: Learning Representation across Arbitrary Graphs

Dooho Lee, Myeong Kong, Minho Jeong, Jaemin Yoo

ICML 2026 | arXiv | GitHub | OpenReview

Generalizing Multi-Scale Time-Series Modeling with a Single Operator

Cheonwoo Lee, **Dooho Lee**, Doyun Choi, Jaemin Yoo

ICML 2026 | arXiv | GitHub | OpenReview

A Probabilistic Circuit Framework for Interpretable Graph PU Learning

Sagad Hamid, **Dooho Lee**, Myeong Kong, Tanya Braun, Jaemin Yoo

UAI 2026 | PMLR | GitHub | OpenReview

Aggregation Buffer: Revisiting DropEdge with a New Parameter Block

Dooho Lee, Myeong Kong, Sagad Hamid, Cheonwoo Lee, Jaemin Yoo

ICML 2025 | PMLR | arXiv | GitHub | OpenReview

Preprints

Distillation of Tabular Foundation Models into Efficient Predictors

Minho Jeong, **Dooho Lee**, Jinmo Lee, Jaemin Yoo

1 Oct 2026 | arXiv

NodeGround: A Node Classification Benchmark in the Graph Foundation Model Era

Jinmo Lee, **Dooho Lee**, Minho Jeong, Jaemin Yoo

30 Sep 2026 | arXiv

Molecular Property Prediction under Structural Shift with Tabular Foundation Models

Jinmo Lee, **Dooho Lee**, Minho Jeong, Jaemin Yoo

30 Sep 2026 | arXiv

Message Passing Does More with Less for In-Context Learning on Graphs

Dooho Lee, Jinmo Lee, Minho Jeong, Kijung Shin, Jaemin Yoo

29 Sep 2026 | arXiv | GitHub | Hugging Face

TaskBridge: Bridging Unsupervised Tabular Anomaly Detection and In-Context Learning via Virtual Tasks

Doyun Choi, **Dooho Lee**, Jaemin Yoo

29 Sep 2026 | arXiv

Causilo Technical Report

Minyong Cho, Minho Jeong, **Dooho Lee**, Jinmo Lee, Jaemin Yoo

19 Sep 2026 | arXiv | GitHub | Hugging Face

Work Experience

Co-Founder and CTO

Nums AI | Seoul, Republic of Korea | Apr 2026 - Present

Co-founded Nums AI to build tabular foundation models for structured data and leads technology strategy and engineering as CTO.

Founding Engineer & Designer

Pensive | Berkeley, CA | Nov 2022 - Aug 2023

Co-founded a US-based startup with two UC Berkeley students and worked across engineering and product design for LLM-powered workflow tools. Built Pensieve Extension for real-time note taking from Chrome highlights and Pensieve Notes for Zoom transcription and automatic meeting notes, which was used by 3 organizations.

Korean Augmentation to the United States Army (KATUSA)

US Army | USAG Humphreys, Republic of Korea | Nov 2021 - May 2023

Served as a KATUSA at Bravo Company, 602nd Aviation Support Battalion, 2nd Combat Aviation Brigade, 2nd Infantry Division. Supported communication and coordination between US and Korean personnel while serving as the only KATUSA in the largest company of the unit.

Founding Engineer & Designer

LayUs | Seoul, Republic of Korea | Mar 2022 - Jul 2022

Built a mobile application for exhibition visitors to upload ticket information and redeem F&B coupons. Contributed to product implementation for a ticket-based redemption workflow connecting cultural events with local customer benefits.

Co-Founder, Software Engineer & Designer

WARD | Daejeon, Republic of Korea | Sep 2020 - Oct 2021

Co-founded and built a machine learning-based stock information service. The team was selected as one of the top 12 teams in 2021 E*5 KAIST under Blue Point Partners mentorship, launched a private beta with 200+ users, and gathered feedback through fund manager sessions, interviews, and surveys.

Industrial Project

Graph-Based Tracking of News Propagation on Social Media Using LLMs

National Cyber Security Research Institute | Republic of Korea | 2025

Worked on an industrial research project for graph-based tracking of news propagation on social media using LLMs.

Teaching Experience

Teaching Assistant, Introduction to Electronics Design Lab

EE300, KAIST | Spring 2026

Teaching Assistant, Seminar<Colloquium>

EE931/932, KAIST | Fall 2026

Teaching Assistant, Programming Structures for Electrical Engineering

EE209, KAIST | Spring/Fall 2025

Developed a data structures programming assignment with auto-grading for 200+ Electrical Engineering majors each semester; assignments/mmr_db.

Teaching Assistant, Foundation of Big Data Analytics

EE412, KAIST | Fall 2025

Invited Talk

Invited Research Talk, Korea Software Congress 2025, Yeosu, Republic of Korea, Dec 2025.

Professional Service

Reviewer, ICLR (2027)
Reviewer, GFM @ ICML 2026
Reviewer, NeurIPS (2026)

Selected Awards & Honors

Sep 2026 Outstanding Master's Student Scholarship, KOSAF (₩2.5M/semester)
Jun 2024 2024 Team KAIST Global Challenge Program, $15,000 funding
Oct 2023 2nd place, 2023 Korea SW Startup Ideathon, ₩1.5M prize
Apr 2023 1st place, 8th Army ROKA Support Group Insignia Competition
Nov 2022 2nd place, Art Service Follow-up Growth Support Project, ₩20M prize
Sep 2021 One of 12 finalists, E*5 KAIST
Apr 2021 1st place, Creative Space G A.I & IoT Hackathon, ₩1M prize